

Unit 3 – Communication Skills for Presentation: Writing, Designing, and Speaking

Thesis and Project Report Writing, Technical Proposal Writing, How to Pitch an Idea: Process, Preparation and Structure, Elements of Speech Delivery: Passion, Poise & Illustrations.

1. Thesis and Project Report Writing

◇ Meaning

- A **Thesis** or **Project Report** is a **formal written document** that explains your **research work, project, or study**.
- It shows what problem you studied, how you solved it, and what results you got.

◇ Purpose

- To present your **technical knowledge, research skills, and communication ability**.
- It is often required in **final year projects** or **postgraduate studies**.

◇ Structure of a Thesis / Project Report

1. **Title Page** – Title, author name, roll number, department, year.
2. **Certificate** – Signed by guide or department head.
3. **Acknowledgement** – Thanking people who helped you.
4. **Abstract** – Short summary (about 200–300 words).
5. **Table of Contents** – List of chapters and page numbers.
6. **Introduction** – Topic background and objective.
7. **Literature Review** – Summary of previous studies or work.
8. **Methodology** – How you did the project (tools, data, steps).
9. **Results & Discussion** – Findings, graphs, and analysis.
10. **Conclusion & Future Scope** – What you achieved and next steps.
11. **References / Bibliography** – Books, papers, or websites used.
12. **Appendices (Optional)** – Extra data, code, or charts.

◇ Tips for Writing

- Use **clear, formal English** (no slang or chat words).
- Use **headings, bullet points, tables, and charts**.
- Write in **past tense** when describing completed work.
- Proofread before submission.

◇ Example (Short Abstract)

“This project aims to develop a crop prediction model using Machine Learning. Using Python libraries like Scikit-learn and Pandas, the model predicts crop yield based on weather and soil data with 92% accuracy.”

2. Technical Proposal Writing

◇ Meaning

- A **Technical Proposal** is a **document that suggests a plan or solution** for a technical problem or project.
- It is written to **convince others** (like a client, company, or college committee) to **approve your idea** or **fund your project**.

◇ Purpose

- To explain **what you plan to do, how, and why it is useful**.
- It's like a **blueprint before doing actual work**.

◇ Structure of a Technical Proposal

1. **Title / Cover Page** – Project title, author, and date.
2. **Executive Summary / Abstract** – Short overview of the idea.
3. **Problem Statement** – What issue are you solving?
4. **Objectives** – What you aim to achieve.
5. **Proposed Method / Solution** – Your plan, tools, and approach.
6. **Budget and Resources** – Time, cost, and materials required.
7. **Timeline / Schedule** – Expected completion stages.
8. **Expected Outcome** – Benefits and results.
9. **Conclusion** – Why your idea should be accepted.

◇ Example

Title: Smart Waste Management System

Problem: Uncontrolled waste disposal in cities.

Solution: IoT-based smart bins that notify when full.

Outcome: Cleaner environment, efficient waste collection.

◇ Tips

- Be **precise, realistic, and professional**.
- Use **diagrams or flowcharts** if needed.
- Avoid copying from the internet — focus on originality.

3. How to Pitch an Idea

◇ Meaning

- An **idea pitch** is a **short, focused presentation** where you **explain your idea** to investors, teachers, or judges to **gain approval or support**.

◇ Purpose

- To **convince** others that your idea is **valuable, practical, and worth investing in**.

◇ Process of Pitching an Idea

1. Preparation Stage

- a. Research your audience (who will listen).
- b. Understand the **problem your idea solves**.
- c. Prepare slides or visuals (3–5 max).

2. Presentation Stage

- a. Speak confidently and clearly.
- b. Focus on **benefits** and **uniqueness** of your idea.
- c. Keep it short — usually **2–5 minutes**.

3. Follow-Up Stage

- a. Be ready for **questions**.
- b. Show **enthusiasm and adaptability**.

◇ Structure of a Good Pitch

1. **Introduction** – Who you are and what the idea is.
2. **Problem** – What issue are you solving?
3. **Solution** – How your idea solves it.
4. **Market/Usefulness** – Who will use it and how it helps.
5. **Call to Action** – Ask for support, funding, or approval.

◇ Example Pitch (Short)

“I’m Gautam, a Data Science student. My idea is a *Smart Attendance System* using face recognition. It reduces manual errors and saves teachers’ time. With low-cost cameras and Python-based recognition, it’s practical for colleges.”

◇ Tips for a Successful Pitch

- **Practice multiple times.**
- Use **simple language** and strong visuals.
- Show **confidence, not overconfidence**.
- Always **end with impact** (“This idea can change how attendance works!”).

4. Elements of Speech Delivery: Passion, Poise, and Illustrations

◇ Meaning

- **Speech delivery** means **how you present your message** — your voice, expressions, and body language.
- A good speaker balances **Passion, Poise, and Illustrations**.

A. Passion

- **Definition:** Speaking with energy, enthusiasm, and emotion.
- **Why it matters:** Keeps the audience interested and shows confidence.
- **How to show passion:**
 - Use a **lively tone** and **eye contact**.
 - Speak about topics you really care about.
 - Use examples or stories to make points.

Example: While presenting “AI in Healthcare,” talk about how AI saves lives — not just data accuracy.



B. Poise

- **Definition:** Staying calm, confident, and graceful under pressure.
- **Why it matters:** Shows professionalism and maturity.
- **How to maintain poise:**
 - Stand straight with relaxed posture.
 - Don’t fidget or move too much.
 - Control your tone and avoid rushing.

- Smile naturally and breathe evenly.

Example: If someone interrupts during your talk, politely pause and respond confidently instead of panicking.

C. Illustrations

- **Definition:** Using **examples, visuals, or stories** to explain complex ideas.
- **Purpose:** Makes your talk more **interesting and easy to remember**.
- **Types of Illustrations:**
 - **Visuals:** Images, graphs, slides.
 - **Verbal Examples:** Real-life stories or analogies.
 - **Demonstrations:** Showing how something works.

Example: When explaining “Neural Networks,” show a simple diagram comparing it to how the human brain works.

Summary Table

Topic	Purpose	Example
Thesis / Project Report	Present research or project work formally.	Final year BTech report on AI project.
Technical Proposal	Suggest plan or solution before starting project.	IoT-based water monitoring proposal.
Idea Pitching	Convince others about your idea in short time.	3-min startup pitch on a smart app.
Speech Delivery	Present ideas confidently and clearly.	Technical presentation or seminar.

Important Difference Tables – Unit 3 (Technical Communication)

1. Thesis vs Project Report

Basis	Thesis	Project Report
Meaning	A research document presenting original study or findings.	A document showing practical work or application of theory.
Purpose	To contribute new knowledge or ideas.	To show technical skills and project implementation.
Focus	Research, theory, and analysis.	Design, execution, and results.
Length	Usually longer (for PG or PhD).	Shorter (for BTech or internships).
Example	“Study of AI Algorithms for Disease Prediction.”	“Development of Crop Prediction App using Python.”

2. Thesis / Project Report vs Technical Proposal

Basis	Thesis / Project Report	Technical Proposal
Purpose	To present completed work or research findings.	To suggest a plan or idea before starting a project.
Time of Writing	After project completion.	Before project begins.
Tone	Informative and analytical.	Persuasive and convincing.
Content	Background, methods, results, conclusion.	Problem, objective, proposed solution, budget.
Example	Final year project report.	Proposal for funding an IoT system.

3. Technical Proposal vs Idea Pitch

Basis	Technical Proposal	Idea Pitch
Mode	Written document.	Spoken/visual presentation.
Purpose	To get approval for a technical project.	To convince audience quickly about the idea’s value.
Length	Detailed and formal (few pages).	Short and direct (2–5 minutes).
Audience	Management, faculty, or investors.	Judges, investors, or panel.
Tone	Formal and data-driven.	Energetic and persuasive.

Example	Written plan for AI-based traffic control.	Verbal pitch for smart traffic app.
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4. Written Communication vs Oral Communication (Presentation)

Basis	Written Communication	Oral Communication
Form	Reports, proposals, emails, theses.	Speeches, lectures, seminars.
Medium	Written (paper or digital).	Spoken (voice, gestures).
Feedback	Delayed.	Immediate.
Record	Permanent record.	Temporary, unless recorded.
Skills Needed	Writing, grammar, structure.	Speaking, tone, body language.
Example	Project report submission.	Seminar presentation.

5. Presentation (Writing) vs Presentation (Speaking)

Basis	Writing for Presentation	Speaking for Presentation
Purpose	To organize content and ideas.	To deliver the message effectively.
Focus	Structure, clarity, visuals.	Voice, expression, and engagement.
Preparation	Script, slides, and handouts.	Practice tone, pace, and gestures.
Evaluation	Based on content.	Based on delivery.
Example	Preparing PPT slides.	Presenting PPT in front of class.

6. Idea Pitch vs Speech

Basis	Idea Pitch	Speech
Objective	To convince or sell an idea quickly.	To inform, motivate, or entertain an audience.
Duration	2–5 minutes.	5–20 minutes or more.
Focus	Problem-solution-benefit.	Message and audience connection.
Tone	Energetic and persuasive.	Depends on purpose (formal or emotional).

Example	Pitching startup idea.	Motivational talk on innovation.
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7. Passion vs Poise (Elements of Speech)

Basis	Passion	Poise
Meaning	Expressing strong enthusiasm and energy while speaking.	Maintaining calmness and confidence during presentation.
Effect on Audience	Builds excitement and interest.	Builds trust and credibility.
Shown By	Voice modulation, expressions, gestures.	Body language, posture, eye contact.
Overuse Risk	Can appear over-emotional.	Can appear dull if excessive.
Example	Speaking emotionally about technology changing lives.	Staying calm during tough questions.

8. Illustrations vs Examples

Basis	Illustrations	Examples
Meaning	Visual or story-based explanation of an idea.	Real or imaginary case used to clarify a point.
Form	Diagrams, stories, charts.	Short cases or facts.
Purpose	To make concept interesting and visual.	To make concept understandable and realistic.
Used In	Speeches, lectures, designs.	Reports, essays, and discussions.
Example	Showing a chart of data flow.	Saying “For example, in AI, algorithms learn from data.”

9. Visual Presentation vs Verbal Presentation

Basis	Visual Presentation	Verbal Presentation
Medium	Slides, charts, infographics.	Spoken words and tone.
Purpose	To support ideas using visuals.	To explain and engage audience.
Tools Used	PowerPoint, Canva, charts.	Voice, gestures, eye contact.

Strength	Improves understanding quickly.	Builds connection and clarity.
Example	Showing a bar chart of results.	Explaining what that chart means.

10. Formal Presentation vs Informal Presentation

Basis	Formal Presentation	Informal Presentation
Setting	Professional (seminar, conference).	Friendly or casual (classroom, club).
Tone	Serious, structured, and polite.	Relaxed and conversational.
Preparation	Planned and rehearsed.	Flexible and spontaneous.
Dress Code	Formal wear.	Casual allowed.
Example	Presenting project to faculty panel.	Sharing project idea with friends.

11. Academic Report vs Business Report

Basis	Academic Report	Business Report
Purpose	To document study or research findings.	To present information for business decisions.
Audience	Teachers, researchers, students.	Managers, clients, stakeholders.
Language Style	Formal, detailed, theoretical.	Professional, practical, concise.
Example	Thesis on “AI Algorithms.”	Report on “AI Market Analysis.”

12. Research Report vs Project Report

Basis	Research Report	Project Report
Objective	To study and analyze data for new findings.	To describe how a project was built or implemented.
Nature	Theoretical and data-driven.	Practical and result-oriented.
Output	Knowledge or conclusion.	Product, model, or application.

Example	Research on climate data trends.	IoT project for smart farming.
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13. Good Speaker vs Poor Speaker

Basis	Good Speaker	Poor Speaker
Voice	Clear and confident.	Dull or shaky.
Body Language	Relaxed and natural.	Nervous or stiff.
Audience Engagement	Maintains eye contact, interacts.	Reads slides without connecting.
Preparation	Well-practiced.	Unprepared.
Example	Delivers technical seminar smoothly.	Reads everything from notes.

14. Passionate Speaker vs Mechanical Speaker

Basis	Passionate Speaker	Mechanical Speaker
Energy Level	High and inspiring.	Low and monotone.
Audience Impact	Motivates listeners.	Bores or loses audience.
Use of Emotions	Natural and expressive.	Lacks emotion or life.
Example	Speaker excited about AI's future.	Speaker reading AI facts dully.

15. Good Proposal vs Poor Proposal

Basis	Good Proposal	Poor Proposal
Clarity	Clear objectives and problem statement.	Vague and confusing.
Structure	Well-organized sections.	Unstructured and messy.
Feasibility	Realistic goals and budget.	Unrealistic expectations.
Language	Formal and simple.	Complex or incorrect.
Impact	Impresses evaluator.	Rejected due to poor communication.